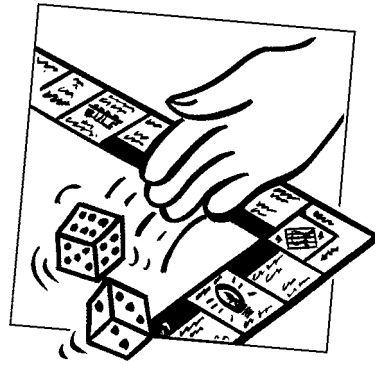


CHAPTER 4

the random behavior of assets



The aim of this Chapter...

... is to demonstrate mathematical modeling in practice, to take the reader from an analysis of stock price data towards a probabilistic model for the behavior of asset prices. The fundamental model we build up will later be used as the starting point for deriving the famous Black–Scholes model for option prices. By the end of this chapter the reader will feel comfortable performing a simple analysis of any financial data.

In this Chapter...

- more notation commonly used in mathematical finance
- how to examine time-series data to model returns
- the Wiener process, a mathematical model of randomness
- a simple model for equities, currencies, commodities and indices

4.1 INTRODUCTION

In the first few chapters you've seen some basic financial instruments, and you've seen some of the important financial concepts such as no arbitrage and hedging. Now we are going to go right back to first principles with data analysis and modeling. We'll also start to see some stochastic calculus. No more of that binomial nonsense, although it gave us some useful insight at the time. (Do you get the sense that I have a mission here?) In this chapter I describe a simple continuous-time model for equities and other financial instruments, inspired by our earlier coin-tossing experiment. This takes us into the world of stochastic calculus and Wiener processes. Although there is a great deal of theory behind the ideas I describe, I am going to explain everything in as simple and accessible manner as possible. We will be modeling the behavior of equities, currencies and commodities, but the ideas are applicable to the fixed-income world as well.

4.2 THE POPULAR FORMS OF 'ANALYSIS'

There are three forms of 'analysis' commonly used in the financial world:

- Fundamental
- Technical
- Quantitative

Fundamental analysis is all about trying to determine the 'correct' worth of a company by an in-depth study of balance sheets, management teams, patent applications, competitors, lawsuits, etc. In other words, getting to the heart of the firm, doing lots of accounting and projections and what-not. This sounds like a really sensible way to model a company and hence its stock price. There are unfortunately two difficulties with this approach. First it is very, very hard. You need a degree in accounting and plenty of patience. And even then all the most important stuff can be hidden 'off balance sheet.' Second, and more importantly, 'The market can stay irrational longer than you can stay solvent' (Keynes). In other words, even if you have the perfect model for the value of a firm it doesn't mean you can make money. You have to find some mispricing and then hope that the rest of the world starts to see your point of view. And this may never happen. If fundamental analysis is hard, then the next form of analysis is the exact opposite, because it is so easy.

Technical analysis is when you don't care anything about the company other than the information contained within its stock price history. You draw trendlines, look for specific patterns in the share price and make predictions accordingly. This is the subject of Appendix B. Most academic evidence suggests that most technical analysis is bunk.

The final form of analysis is the one we are really concerned with in this book, and is the form that has been most successful over the last 50 years, forming a solid foundation for portfolio theory, derivatives pricing and risk management. It is **quantitative analysis**. Quantitative analysis is all about treating financial quantities such as stock prices or interest rates as random, and then choosing the best models for that randomness. Let's see why randomness is important and then build up a simple, random, stock price model.

4.3 WHY WE NEED A MODEL FOR RANDOMNESS: JENSEN'S INEQUALITY

Why is 'randomness' so crucial to modeling the world of derivatives? Why can't we just try to forecast the future stock price as best we can and figure out the option's payoff? To best see the importance of randomness in option theory let's take a look at some very simple mathematics, called **Jensen's inequality**.

The stock price today is 100. Let's suppose that in one year's time it could be 50 or 150, with both equally likely. See Figure 4.1. How can we value an option on this stock, a call option with a strike of 100 expiring in one year, say?

Two ways spring to mind.

With those two possible scenarios we could say that we expect the stock price to be at 100 in one year, this being the average of the possible future values. The payoff for the call option would then be 0, since it is exactly at the money. And the present value of this is zero. Could this be the way to value an option?

Probably not. You expect the value to be greater than zero, since half the time there is some payoff.

Alternatively we could look at the two possible payoffs and then calculate that expectation. If the stock falls to 50 then the payoff is zero, if it rises to 150 then the payoff is 50.

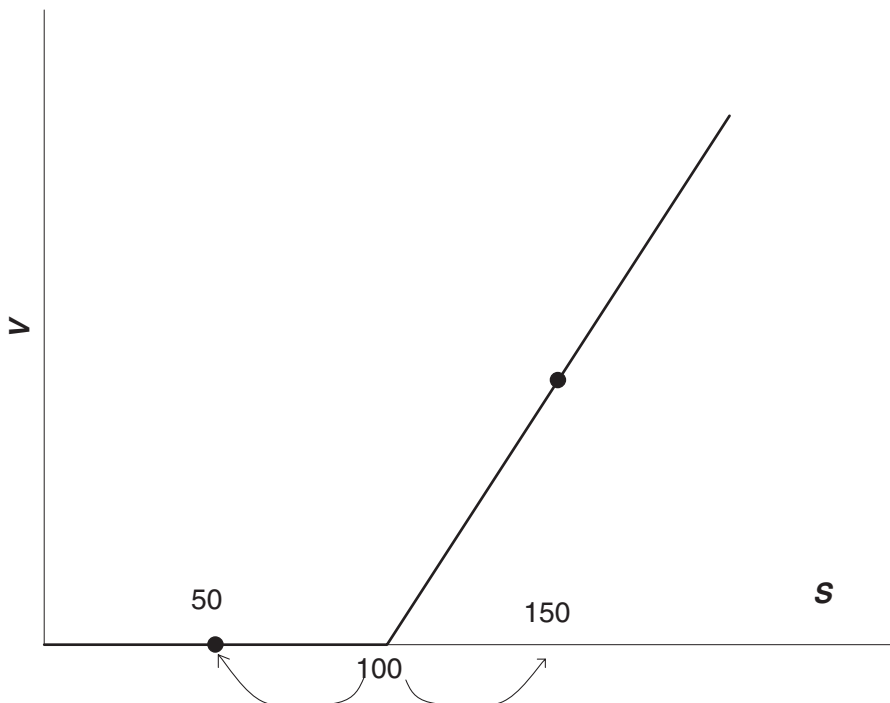


Figure 4.1 Future scenarios.

The average payoff is therefore 25, which we could present value to give us some idea of the option's value.

It turns out that the second calculation is closer to what we do in practice to value options (although we know that the *real* probabilities don't come into the calculation) and we'll see lots of this throughout the book. But that calculation also illustrates another point of great importance, that the order in which we do the payoff calculation and the expectation matters. In this example we had

$$\text{Payoff (Expected [Stock price])} = 0$$

whereas

$$\text{Expected [Payoff(Stock price)]} = 25.$$

This is an example of Jensen's inequality. Let's use some symbols. If we have a convex function $f(S)$ (in our example the payoff function for a call) of a random variable S (in our example the stock price) then

$$E[f(S)] \geq f(E[S]). \quad (4.1)$$

We can even get an idea of how much greater the left-hand side is than the right-hand side by using a Taylor series approximation around the mean of S . Write

$$S = \bar{S} + \epsilon,$$

where $\bar{S} = E[S]$, so the $E[\epsilon] = 0$. Then

$$\begin{aligned} E[f(S)] &= E[f(\bar{S} + \epsilon)] = E\left[f(\bar{S}) + \epsilon f'(\bar{S}) + \frac{1}{2}\epsilon^2 f''(\bar{S}) + \dots\right] \\ &\approx f(\bar{S}) + \frac{1}{2}f''(\bar{S})E[\epsilon^2] \\ &= f(E[S]) + \frac{1}{2}f''(E[S])E[\epsilon^2]. \end{aligned}$$

So the left-hand side of (4.1) is greater than the right by approximately

$$\frac{1}{2}f''(E[S])E[\epsilon^2].$$

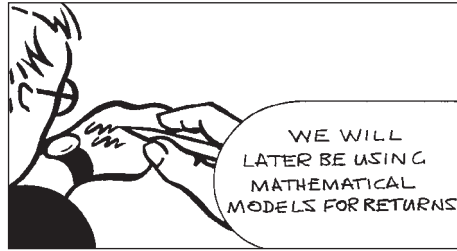
This shows the importance of two concepts:

- $f''(E[S])$: The **convexity** of an option. As a rule this adds value to an option. It also means that any intuition we may get from linear contracts (forwards and futures) might not be helpful with non-linear instruments such as options.
- $E[\epsilon^2]$: Randomness in the underlying, and its **variance**. As stated above, modeling randomness is the key to modeling options.

Now we have seen a hint as to why randomness is so important, let's start modeling some assets! The new model we'll be seeing for assets is not unlike the binomial model just described, just a lot more realistic.

4.4 **SIMILARITIES BETWEEN EQUITIES, CURRENCIES, COMMODITIES AND INDICES**

When you invest in something, whether it is a stock, commodity, work of art or a racehorse, your main concern is that you will make a comfortable return on your investment. By **return** we tend to mean the percentage growth in the value of an asset, together with accumulated dividends, over some period:



$$\text{Return} = \frac{\text{Change in value of the asset} + \text{accumulated cashflows}}{\text{Original value of the asset}}.$$

I want to distinguish here between the percentage or relative growth and the absolute growth. Suppose we could invest in either of two stocks, both of which grow on average by \$10 *per annum*. Stock A has a value of \$100 and stock B is currently worth \$1000. Clearly the former is a better investment, at the end of the year stock A will probably be worth around \$110 (if the past is anything to go by) and stock B \$1010. Both have gone up by \$10, but A has risen by 10% and B by only 1%. If we have \$1000 to invest we would be better off investing in ten of asset A than one of asset B. This illustrates that when we come to model assets, it is the return that we should concentrate on. In this respect, all of equities, currencies, commodities and stock market indices can be treated similarly. What return do we expect to get from them?

Part of the business of estimating returns for each asset is to estimate how much unpredictability there is in the asset value. In the next section I am going to show that randomness plays a large part in financial markets, and start to build up a model for asset returns incorporating this randomness.

Time Out...

Returns

Here is another way of understanding why returns are more important than actual stock price. Suppose I told you that one stock had a value of 5 and another 500.

You would think nothing of it. Now suppose I told you that one currency had an interest rate of 5% and another had an interest rate of 500%. Whoa...you'd be somewhat surprised by the currency with the 500% interest rate, wouldn't you? There's a big clue in such an observation. We don't care about the absolute value of a stock price, only its return. So let's analyze and model returns. When we come to modeling interest rates we won't have such a valuable clue to help us. This makes interest rate modeling harder than equity modeling.





On a spreadsheet, I include some data for you to play with

4.5 EXAMINING RETURNS

In Figure 4.2 I show the quoted price of Perez Companc, an Argentinian conglomerate, over the period February 1995 to November 1996. This is a very typical plot of a financial asset. The asset shows a general upward trend over the period but this is far from guaranteed. If you bought and sold at the wrong times you would lose a lot of money. The unpredictability that is seen in this figure is the main feature of financial modeling. Because there is so much randomness, any mathematical model of a financial asset must acknowledge the randomness and have a probabilistic foundation.

Remembering that the returns are more important to us than the absolute level of the asset price, I show in Figure 4.3 how to calculate returns on a spreadsheet. Denoting the asset value on the i th day by S_i , then the return from day i to day $i + 1$ is given by

$$\frac{S_{i+1} - S_i}{S_i} = R_i.$$

(I've ignored dividends here, they are easily allowed for, especially since they only get paid two or four times a year typically.) Of course, I didn't need to use data spaced at intervals of a day, I will comment on this later.

In Figure 4.4 I show the daily returns for Perez Companc. This looks very much like 'noise,' and that is exactly how we are going to model it. The mean of the returns



Figure 4.2 Perez Companc from February 1995 to November 1996.

Date	Perez	Return			
01-Mar-95	2.11		Average return	0.002916	
02-Mar-95	1.90	-0.1	Standard deviation	0.024521	
03-Mar-95	2.18	0.149906			
06-Mar-95	2.16	-0.01081			
07-Mar-95	1.91	-0.11258	=AVERAGE(C3:C463)		
08-Mar-95	1.86	-0.02985			
09-Mar-95	1.97	0.061538			
10-Mar-95	2.27	0.15	=STDEVP(C3:C463)		
13-Mar-95	2.49	0.099874			
14-Mar-95	2.76	0.108565			
15-Mar-95	2.61	-0.05426			
16-Mar-95	2.67	0.021858			
17-Mar-95	2.64	-0.0107			
20-Mar-95	2.60	-0.01622	=(B13-B12)/B12		
21-Mar-95	2.59	-0.00275			
22-Mar-95	2.59	-0.00275			
23-Mar-95	2.55	-0.01232			
24-Mar-95	2.73	0.069307			
27-Mar-95	2.91	0.064815			
28-Mar-95	2.92	0.002899			
29-Mar-95	2.92	0			
30-Mar-95	3.12	0.069364			
31-Mar-95	3.14	0.005405			
03-Apr-95	3.13	-0.00269			
04-Apr-95	3.24	0.037736			
05-Apr-95	3.25	0.002597			
06-Apr-95	3.28	0.007772			
07-Apr-95	3.21	-0.02057			
10-Apr-95	3.02	-0.06037			
11-Apr-95	3.08	0.019553			
12-Apr-95	3.19	0.035616			
17-Apr-95	3.21	0.007936			
18-Apr-95	3.17	-0.01312			
19-Apr-95	3.24	0.021277			

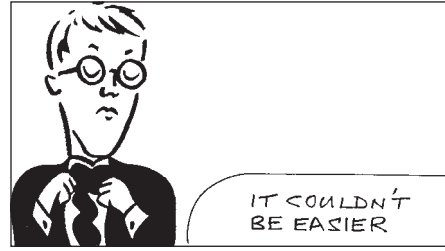


Figure 4.3 Spreadsheet for calculating asset returns.

distribution is

$$\bar{R} = \frac{1}{M} \sum_{i=1}^M R_i \quad (4.2)$$

and the sample standard deviation is

$$\sqrt{\frac{1}{M-1} \sum_{i=1}^M (R_i - \bar{R})^2}, \quad (4.3)$$

where M is the number of returns in the sample (one fewer than the number of asset prices). From the data in this example we find that the mean is 0.002916 and the standard deviation is 0.024521.

Notice how the mean daily return is much smaller than the standard deviation. This is very typical of financial quantities over short timescales. On a day-by-day basis you will tend to see the noise in the stock price, and will have to wait months perhaps before you can spot the trend.

The frequency distribution of this time series of daily returns is easily calculated, and very instructive to plot. In Excel use Tools | Data Analysis | Histogram. In Figure 4.5 is

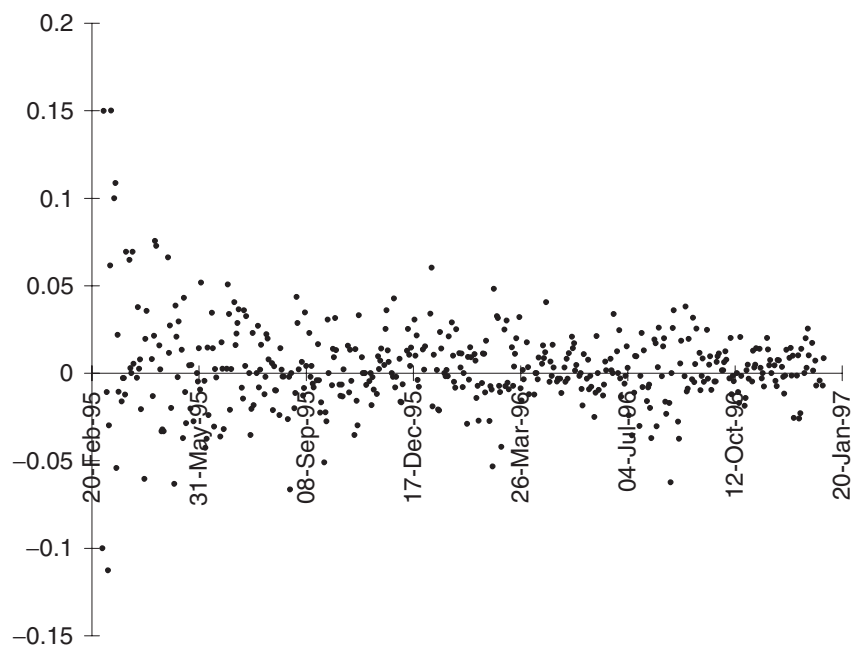


Figure 4.4 Daily returns of Perez Companc.

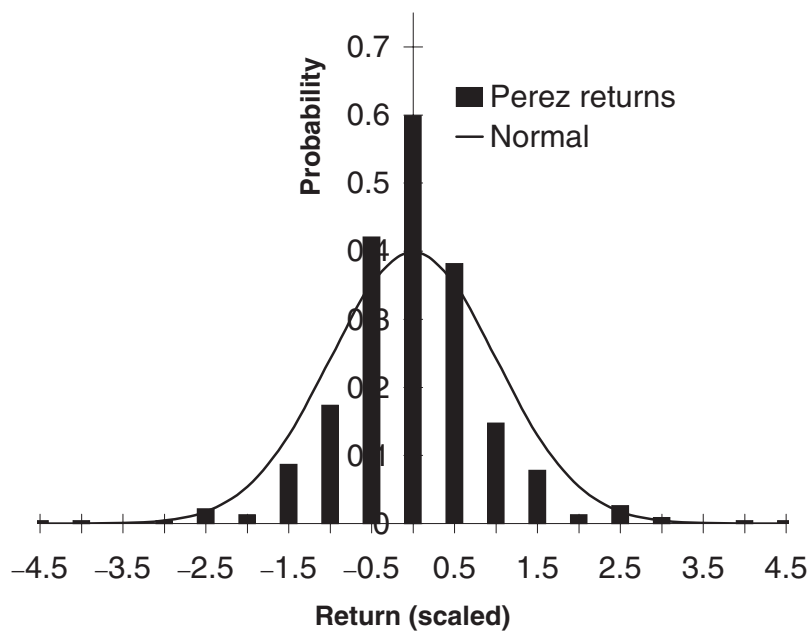


Figure 4.5 Normalized frequency distribution of Perez Companc and the standardized Normal distribution.

shown the frequency distribution of daily returns for Perez Companc. This distribution has been scaled and translated to give it a mean of zero, a standard deviation of one and an area under the curve of one. On the same plot is drawn the probability density function for the standardized Normal distribution function

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\phi^2},$$

where ϕ is a standardized Normal variable.¹ The two curves are not identical but are fairly close.

Supposing that we believe that the empirical returns are close enough to Normal for this to be a good approximation, then we have come a long way towards a model. I am going to write the returns as a random variable, drawn from a Normal distribution with a known, constant, non-zero mean and a known, constant, non-zero standard deviation:

$$R_i = \frac{S_{i+1} - S_i}{S_i} = \text{mean} + \text{standard deviation} \times \phi.$$

Figure 4.6 shows the returns distribution of Glaxo–Wellcome as calculated by Bloomberg. This has not been normalized.

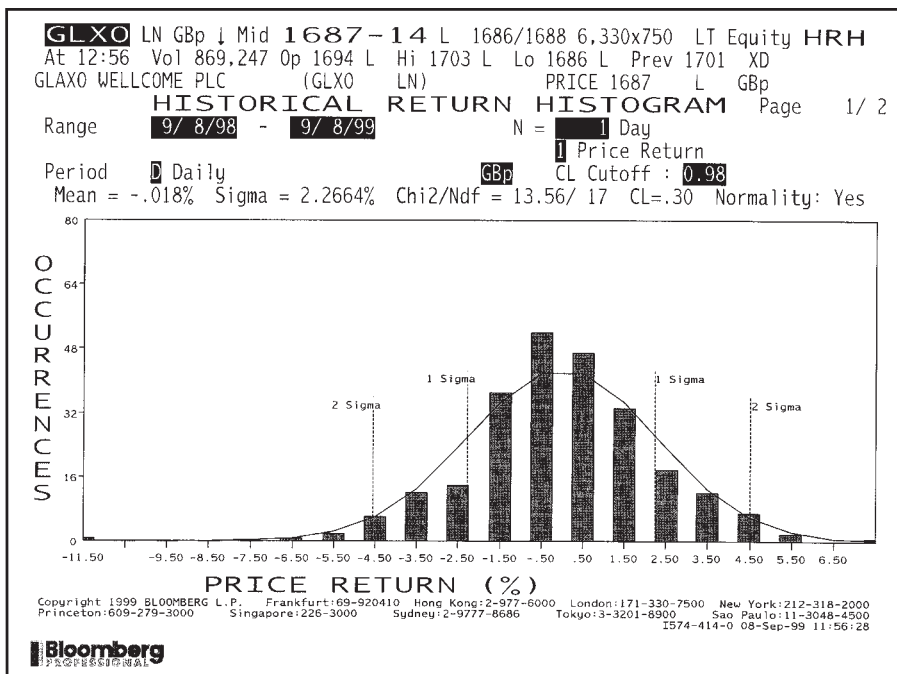


Figure 4.6 Glaxo–Wellcome returns histogram. Source: Bloomberg L.P.

¹ Think of a number, any number ... funny, that's the number I was thinking of.



Time Out...

The Normal distribution

In Excel the bell-shaped standardized Normal probability density function curve is, as a function of X ,

$$1/\text{SQRT}(2*\text{PI}())*\text{EXP}(-0.5*X*X).$$

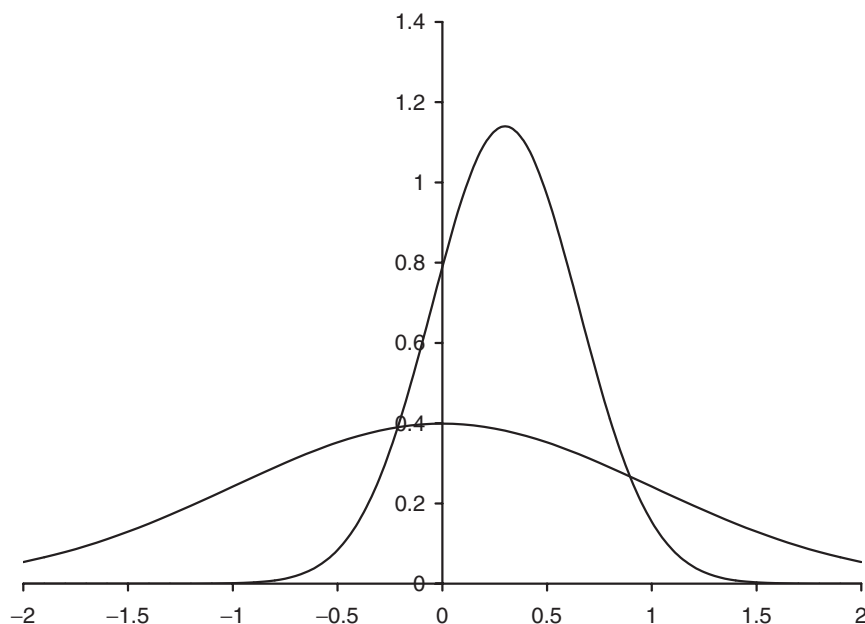
Mathematically this is denoted by $N(0, 1)$. The Normal distribution having mean m and standard deviation s is denoted by $N(m, s^2)$ and in Excel is

$$1/\text{SQRT}(2*\text{PI}())/S*\text{EXP}(-0.5*(X-M)*(X-M)/S/S).$$

The figure below shows a couple of Normal distributions, one is the standardized, and the other has a positive mean and quite a small standard deviation.



The Normal distribution



There are a couple more Time Outs on the Normal distribution in this chapter.

4.6 TIMESCALES

How do the mean and standard deviation of the returns' time series, as estimated by (4.2) and (4.3), scale with the time step between asset price measurements? In the example the time step is one day, but suppose I sampled at hourly intervals or weekly, how would this affect the distribution?

Call the time step δt . The mean of the return scales with the size of the time step. That is, the larger the time between sampling the more the asset will have moved in the meantime, *on average*. I can write

$$\text{mean} = \mu \delta t,$$

for some μ which we will assume to be constant. This is the same μ as in Chapter 3, representing the annualized average return or the drift.

Ignoring randomness for the moment, our model is simply

$$\frac{S_{i+1} - S_i}{S_i} = \mu \delta t.$$

Rearranging, we get

$$S_{i+1} = S_i(1 + \mu \delta t).$$

If the asset begins at S_0 at time $t = 0$ then after one time step $t = \delta t$ and

$$S_1 = S_0(1 + \mu \delta t).$$

After two time steps $t = 2 \delta t$ and

$$S_2 = S_1(1 + \mu \delta t) = S_0(1 + \mu \delta t)^2,$$

and after M time steps $t = M \delta t = T$ and

$$S_M = S_0(1 + \mu \delta t)^M.$$

This is just

$$S_M = S_0 (1 + \mu \delta t)^M = S_0 e^{M \log(1 + \mu \delta t)} \approx S_0 e^{\mu M \delta t} = S_0 e^{\mu T}.$$

In the limit as the time step tends to zero with the total time T fixed, this approximation becomes exact. This result is important for two reasons.

First, in the absence of any randomness the asset exhibits exponential growth, just like cash in the bank.

Second, the model is meaningful in the limit as the time step tends to zero. If I had chosen to scale the mean of the returns distribution with any other power of δt it would have resulted in either a trivial model ($S_T = S_0$) or infinite values for the asset.

The second point can guide us in the choice of scaling for the random component of the return. How does the standard deviation of the return scale with the time step δt ? Again, consider what happens after $T/\delta t$ time steps each of size δt (i.e. after a total time of T). Inside the square root in expression (4.3) there are a large number of terms,

$T/\delta t$ of them. In order for the standard deviation to remain finite as we let δt tend to zero, the individual terms in the expression must each be of $O(\delta t)$. Since each term is a square of a return, the standard deviation of the asset return over a time step δt must be $O(\delta t^{1/2})$:

$$\text{standard deviation} = \sigma \delta t^{1/2},$$

where σ is some parameter measuring the amount of randomness, the larger this parameter the more uncertain is the return. This σ is the same σ we saw in Chapter 3. It is the annualized standard deviation of asset returns.

Putting these scalings explicitly into our asset return model

$$R_i = \frac{S_{i+1} - S_i}{S_i} = \mu \delta t + \sigma \phi \delta t^{1/2}. \quad (4.4)$$

I can rewrite Equation (4.4) as

$$S_{i+1} - S_i = \mu S_i \delta t + \sigma S_i \phi \delta t^{1/2}. \quad (4.5)$$

The left-hand side of this equation is the change in the asset price from time step i to time step $i + 1$. The right-hand side is the ‘model.’ We can think of this equation as a model for a **random walk** of the asset price. This is shown schematically in Figure 4.7. We know exactly where the asset price is today but tomorrow’s value is unknown. It is distributed about today’s value according to (4.5).

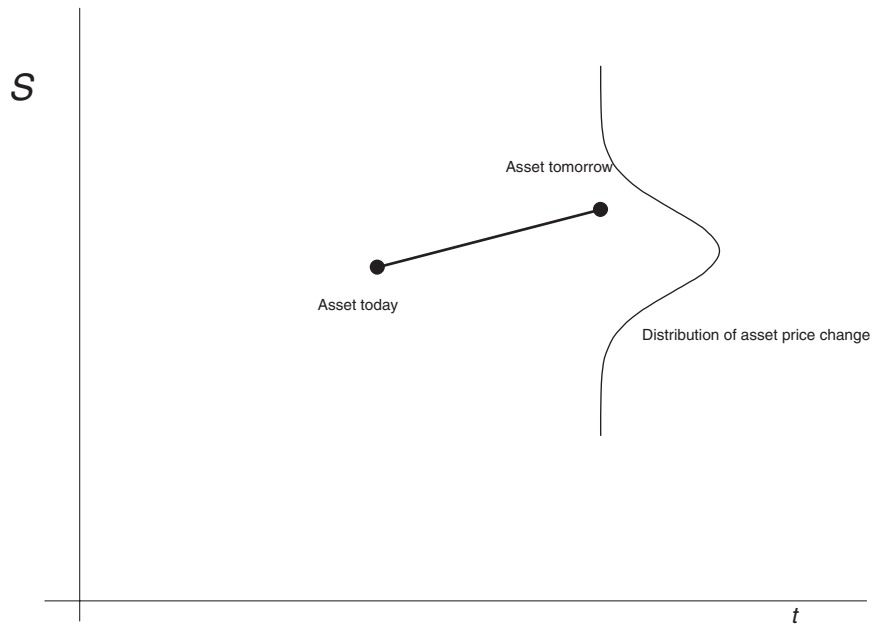
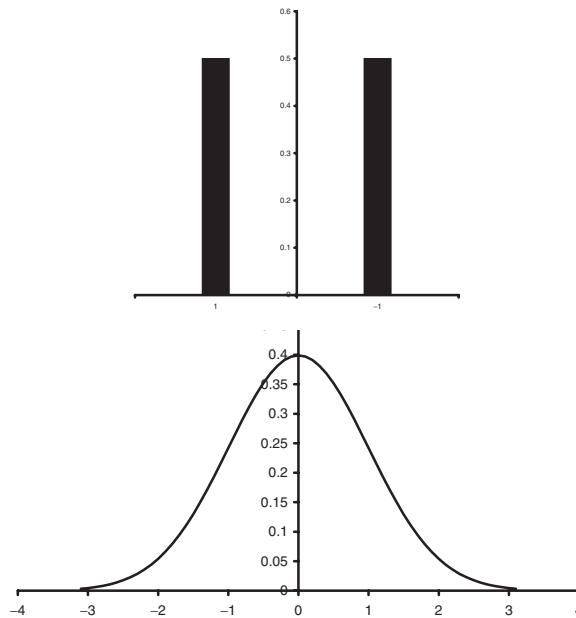


Figure 4.7 A representation of the random walk.

Time Out...

Binomial versus Normal

We've been considering two models for the asset return, the binomial and the Normal. The figure below shows what these two look like. Although completely different, they both should have the same mean return and standard deviation as the asset we are modeling.



4.6.1 The drift

The parameter μ is called the **drift rate**, the **expected return** or the **growth rate** of the asset. Statistically it is very hard to measure since the mean scales with the usually small parameter δt . It can be estimated by

$$\mu = \frac{1}{M \delta t} \sum_{i=1}^M R_i.$$

The unit of time that is usually used is the year, in which case μ is quoted as an *annualized* growth rate.

In the classical option pricing theory the drift plays almost no role. So even though it is hard to measure, this doesn't matter too much.²

4.6.2 The volatility

The parameter σ is called the **volatility** of the asset. It can be estimated by

$$\sqrt{\frac{1}{(M-1)\delta t} \sum_{i=1}^M (R_i - \bar{R})^2}.$$

Again, this is almost always quoted in annualized terms.

The volatility is the most important and elusive quantity in the theory of derivatives. I will come back again and again to its estimation and modeling.

Because of their scaling with time, the drift and volatility have different effects on the asset path. The drift is not apparent over short timescales for which the volatility dominates. Over long timescales, for instance decades, the drift becomes important. Figure 4.8 is a realized path of the logarithm of an asset, together with its expected path and a 'confidence interval.' In this example the confidence interval represents one standard deviation. With the assumption of Normality this means that 68% of the time the asset should be within this range. The mean path is growing linearly in time and the confidence interval grows like the square root of time. Thus over short timescales the volatility dominates.

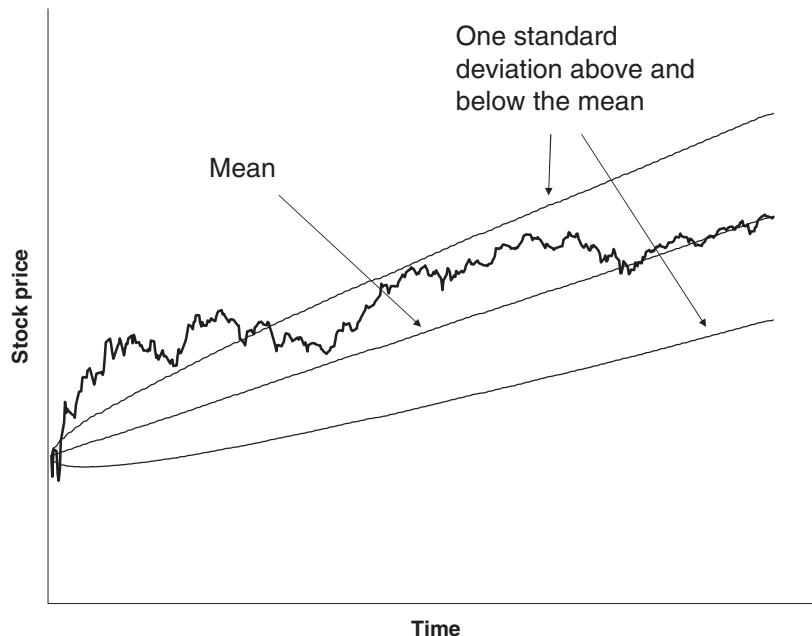


Figure 4.8 Path of the logarithm of an asset, its expected path and one standard deviation above and below.

² In non-classical theories and in portfolio management, it *does* often matter, very much.

4.7 ESTIMATING VOLATILITY

The most common estimate of volatility is simply

$$\sqrt{\frac{1}{(M-1)\delta t} \sum_{i=1}^M (R_i - \bar{R})^2}.$$

If δt is sufficiently small the mean return $\bar{R}$ term can be ignored. For small δt

$$\sqrt{\frac{1}{(M-1)\delta t} \sum_{i=1}^M (\log S(t_i) - \log S(t_{i-1}))^2}$$

can also be used, where $S(t_i)$ is the closing price on day t_i .

It is highly unlikely that volatility is constant for any given asset. Changing economic circumstances, seasonality, etc. will inevitably result in volatility changing with time. If you want to know the volatility today you must use some past data in the calculation. Unfortunately, this means that there is no guarantee that you are actually calculating *today's* volatility.

Typically you would use daily closing prices to work out daily returns and then use the past 10, 30, 100, ... daily returns in the formula above. Or you could use returns over longer or shorter periods. Since all returns are equally weighted, while they are in the estimate of volatility, any large return will stay in the estimate of vol until the 10 (or 30 or 100) days have past. This gives rise to a plateauing of volatility, and is totally spurious.

4.8 THE RANDOM WALK ON A SPREADSHEET

The random walk (4.5) can be written as a 'recipe' for generating S_{i+1} from S_i :

$$S_{i+1} = S_i \left(1 + \mu \delta t + \sigma \phi \delta t^{1/2} \right). \quad (4.6)$$

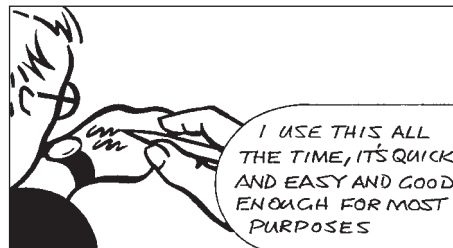
We can easily simulate the model using a spreadsheet. In this simulation we must input several parameters, a starting value for the asset, a time step δt , the drift rate μ , the volatility σ and the total number of time steps. Then, at each time step, we must choose a random number ϕ from a Normal distribution. I will talk about simulations in depth in Chapter 29, for the moment let me just say that an approximation to a Normal variable that is fast in a spreadsheet, and quite accurate, is simply to add up twelve random variables drawn from a uniform distribution over zero to one, and subtract six:

$$\left(\sum_{i=1}^{12} \text{RAND}() \right) - 6.$$

The Excel spreadsheet function `RAND()` gives a uniformly distributed random variable.



This is very simple to implement



	A	B	C	D	E	F	G	H
1	Asset	100		Time	Asset			
2	Drift	0.15		0	100			
3	Volatility	0.25		0.01	98.38844			
4	Timestep	0.01		0.02	94.28005			
5				0.03	95.40441			
6				0.04	92.79735			
7				0.05	93.45168			
8				0.06	93.99664			
9				0.07	97.66597			
10				0.08	96.52319			
11	=E7*(1+\$B\$2*\$B\$4+\$B\$3*SQRT(\$B\$4)*(RAND()+RAND()+RAND()+RAND()+RAND()+RAND()+RAND()+RAND()+RAND()+RAND()+RAND()+RAND()-6))							
12								
13				0.11	99.60075			
14				0.12	99.01974			
15				0.13	100.8729			
16				0.14	101.2378			
17				0.15	102.4736			
18				0.16	102.7694			
19				0.17	100.7347			
20				0.18	102.7021			
21				0.19	107.3493			
22				0.2	109.887			
23				0.21	108.688			
24				0.22	110.7826			
25				0.23	112.8932			
26				0.24	111.0625			
27				0.25	111.6157			
28				0.26	112.5443			
29				0.27	111.9805			
30				0.28	115.6002			
31				0.29	117.9831			
32				0.3	115.2694			
33				0.31	117.4374			

Figure 4.9 Simulating the random walk on a spreadsheet.

In Figure 4.9 I show the details of a spreadsheet used for simulating the asset price random walk.

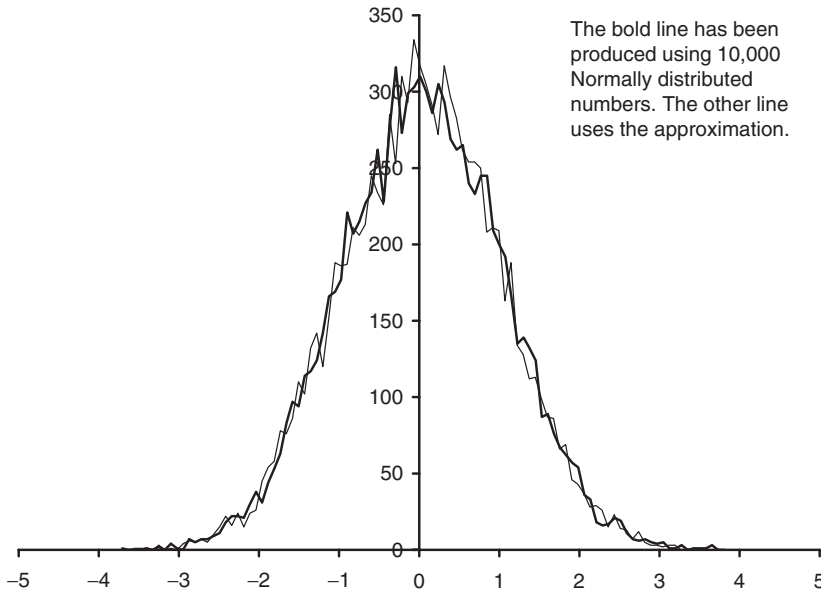


Time Out...

Excel and the approximation to the Normal

You can draw Normally distributed random numbers in Excel using NORMSINV(RAND()). But this is very slow.

Below are the distributions for the real Normal and the approximate Normal using 10,000 random numbers. Not bad?



The bold line has been produced using 10,000 Normally distributed numbers. The other line uses the approximation.



Useful approximations

Why use 12 RANDs? Well, you can use any number, the more you use the closer the approximation will be to Normal, but the longer it will take to compute. The general formula using N uniformly distributed numbers is

$$\sqrt{\frac{12}{N}} \left(\left(\sum_{i=1}^N \text{RAND}() \right) - \frac{N}{2} \right).$$

This has been scaled to have a mean of zero and unit standard deviation.

4.9 THE WIENER PROCESS

So far we have a model that allows the asset to take any value after a time step. This is a step forward but we have still not reached our goal of continuous time, we still have a discrete time step. This section is a brief introduction to the continuous-time limit of equations like (4.4). I will start to introduce ideas from the world of stochastic modeling and Wiener processes, delving more deeply in Chapter 5.

I am now going to use the notation $d\cdot$ to mean ‘the change in’ some quantity. Thus dS is the ‘change in the asset price.’ But this change will be in *continuous time*. Thus we will go to the limit $\delta t = 0$. The first δt on the right-hand side of (4.5) becomes dt but the second term is more complicated.

I cannot straightforwardly write $dt^{1/2}$ instead of $\delta t^{1/2}$. If I do go to the zero-time step limit then any random $dt^{1/2}$ term will dominate any deterministic dt term. Yet in our problem

the factor in front of $dt^{1/2}$ has a mean of zero, so maybe it does not outweigh the drift after all. Clearly something subtle is happening in the limit.

It turns out, and we will see this in Chapter 5, that because the *variance* of the random term is $O(\delta t)$ we can make a sensible continuous-time limit of our discrete-time model. This brings us into the world of Wiener processes.



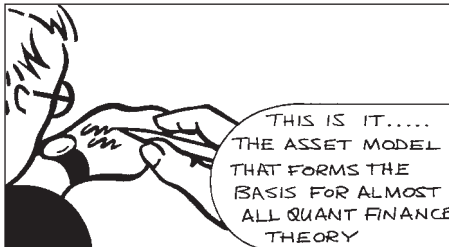
I am going to write the term $\phi \delta t^{1/2}$ as

$$dX.$$

You can think of dX as being a random variable, drawn from a Normal distribution with mean zero and variance dt :

$$E[dX] = 0 \quad \text{and} \quad E[dX^2] = dt.$$

This is not exactly what it is, but it is close enough to give the right idea. This is called a **Wiener process**. The important point is that we can build up a continuous-time theory using Wiener processes instead of Normal distributions and discrete time.



4.10 THE WIDELY ACCEPTED MODEL FOR EQUITIES, CURRENCIES, COMMODITIES AND INDICES

Our asset price model in the continuous-time limit, using the Wiener process notation, can be written as

$$dS = \mu S dt + \sigma S dX. \quad (4.7)$$

This is our first **stochastic differential equation**. It is a continuous-time model of an asset price. It is the most widely accepted model for equities, currencies, commodities and indices, and the foundation of so much finance theory.



Time Out...

Don't panic!

Stochastic differential equations can be a bit unnerving, especially to anyone used to ordinary or partial differential calculus. But do not worry, keep thinking in terms of simulations and the algorithm we've built up in the spreadsheet.

We've now built up a simple model for equities that we are going to be using quite a lot. You could ask, if the stock market is so random how can fund managers justify their fee? Do

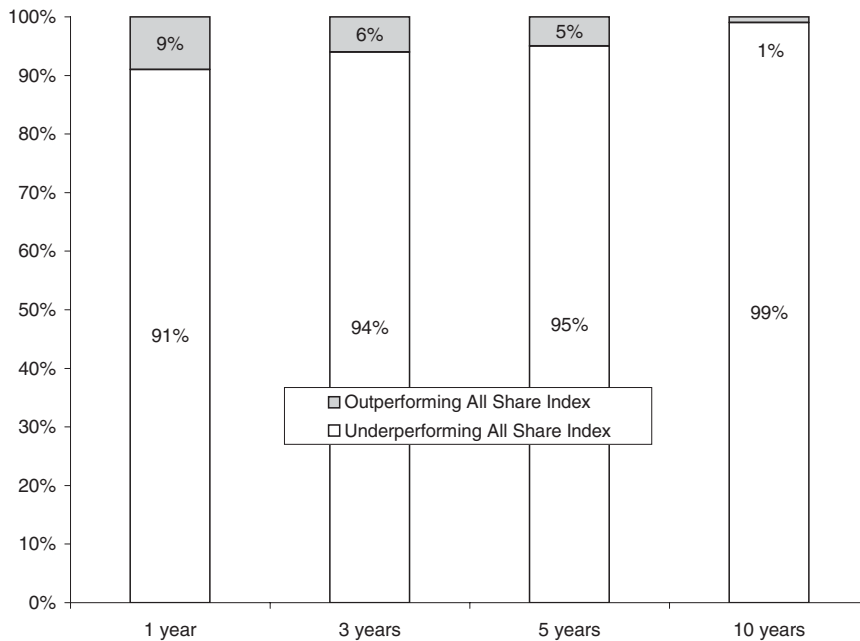


Figure 4.10 Fund performances compared with UK All Share Index. To end December 1998. Data supplied by Virgin Direct.

they manage to outsmart the market? Are they clairvoyant or aren't the markets random? Well, I won't swear that markets are random but I can say with confidence that fund managers don't outperform the market. In Figure 4.10 is shown the percentage of funds that outperform an index of all UK stocks. Whether we look at a one-, three-, five- or 10-year horizon we can see that the vast majority of funds can't even keep up with the market. And statistically speaking, there are bound to be a few that beat the market, but only by chance. Maybe one should invest in a fund that does the opposite of all other funds. Great idea except that the management fee and transaction costs probably mean that that would be a poor investment too. This doesn't prove that markets are random, but it's sufficiently suggestive that most of my personal share exposure is via an index-tracker fund.

Time Out...

Why do we like the Normal distribution?

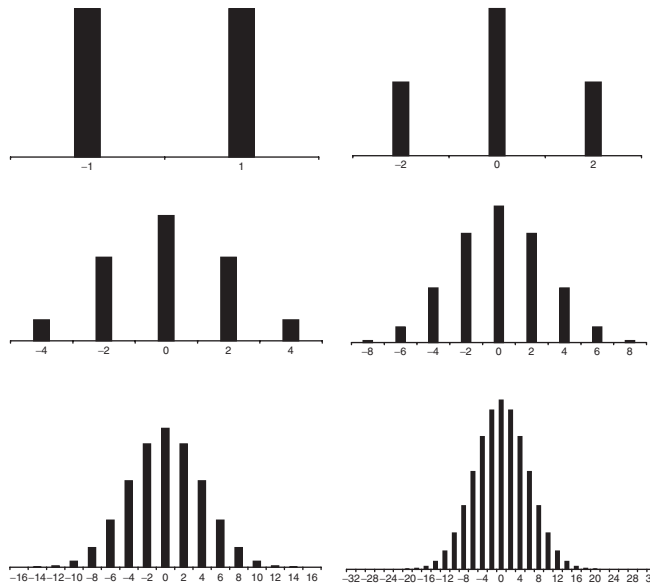
The Normal distribution is a special and wonderful distribution. It occurs naturally in many walks of life, and has nice properties. Here's a little experiment that shows why it crops up naturally, and after this I'll give you a theorem.



Back to coin tossing. Toss one coin, heads you win one dollar, tails you lose one dollar. Figure 1 shows the probability distribution.

Now toss two coins, same rules, for each head you get one dollar, but lose one for each tail. The probability density function is shown in Figure 2.

The sequence of figures below shows the probability density function of winnings/losses after an increasing number of tosses. What do you notice? It's starting to look more and more like the bell-shaped Normal distribution.



This is a simple demonstration of the **Central Limit Theorem**: Let $X_1, X_2, \dots$ be a sequence of independent identically distributed (i.i.d.) random variables with finite means m and finite non-zero variances s^2 then the sum

$$S_n = X_1 + X_2 + \dots + X_n$$

in the limit as $n \rightarrow \infty$ is distributed Normally with mean nm and variance ns^2 . Or if we rescale,

$$S'_n = \frac{X_1 + X_2 + \dots + X_n - nm}{\sqrt{ns}}$$

tends to the standardized Normal distribution.

The point is that if we add up enough i.i.d. random variables (with finite mean and standard deviation) we end up with something that's Normally distributed. And that's why the Normal distribution occurs all over the place.

4.11 SUMMARY

In this chapter I introduced a simple model for the random walk of asset. Initially I built the model up in discrete time, showing what the various terms mean, how they scale with the time step and showing how to implement the model on a spreadsheet.

Most of this book is about continuous-time models for assets. The continuous-time version of the random walk involves concepts such as stochastic calculus and Wiener processes. I introduced these briefly in this chapter and will now go on to explain the underlying theory of stochastic calculus to give the necessary background for the rest of the book.

FURTHER READING

- Mandelbrot (1963) and Fama (1965) did some of the early work on the analysis of financial data.
- Parkinson (1980) derived the high-low estimator and Garman & Klass (1980) derived the high-low-close estimator.
- For an introduction to random walks and Wiener processes see Øksendal (1992) and Schuss (1980).
- Some high frequency data can be ordered through Olsen Associates, www.olsen.ch. It's not free, but nor is it expensive.
- The famous book by Malkiel (1990) is well worth reading for its insights into the behavior of the stock market. Read what he has to say about chimpanzees, blindfolds and darts. In fact, if you haven't already got Malkiel's book make sure that it is the next book you read after finishing mine.

EXERCISES

1. A share has an expected return of 12% per annum (with continuous compounding) and a volatility of 20% per annum. Changes in the share price satisfy $dS = \mu S dt + \sigma S dX$. Simulate the movement of the share price, currently \$100, over a year, using a time interval of one week.
2. What is the distribution of the price increase for the share movement described in Question 1?
3. Using daily share price data, find and plot returns for the asset. What are the mean and standard deviation for the sample you have chosen?
4. Compare interest rate data with your share price data. Are there any major differences? Is the asset price model

$$dS = \mu S dt + \sigma S dX$$

also suitable for modeling interest rates?

